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CIS 344 Exam 1 Take Home
Written version to answers

1. Is there a difference between the average income in the Louisville area for those who are under the age of 50 when compared to those who are 50 and over? Report an estimate of the difference between the means of the two groups and discuss whether the mean values are the same.

First, in order to determine what test to run to figure out if there was a difference in mean incomes for people younger than 50 and 50 or older, I had to run an F-test to determine equal or unequal variance. At the 95% confidence level, with a p-value less than alpha (0.0012< 0.05), we could conclude that the two samples had unequal variance. This told me to run a two-sample t test with independent samples for unequal variance.

The mean income for people under 50 was $57,633.47 and the mean income for people 50 or older was $69,788.86. (Difference of $12,155.39).

At the 95% confidence level, the absolute value of tSTAT > tCRIT (9.87 > 1.9 and a p-value < alpha (0.000 < 0.05), we reject the null hypothesis and conclude the two population mean incomes are different between those younger than 50 and those 50 or older.

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2. What is your best estimate for the proportion of high school dropouts (EDUC < 12) in the Louisville area? Is your estimate consistent with a national average of 8.05%?

This is a 1 sample z test for proportions with a given value. The given value was 189 people in the sample that had less than 12 credit hours. (189/1890 = 0.10)

After setting up and running the z-test, at the 95% confidence level, p-value < alpha (0.002 < 0.05). Therefore, we reject the null hypothesis and conclude that the estimated dropout rate in Louisville is not consistent with the national average of 8.05% with an estimate of 0.1 +/- 0.014 (0.087, 0.113). I am 95% confident that the true proportion of Louisville dropouts lies within this interval.

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3. Is there a difference between the proportion of high school dropouts who are under 35 and those who are over 65? Report an estimate of the di↵erence in the proportions of high school dropouts for the two age groups.

This is a 2 sample z test for proportions. First, I had to get the proportion of high school dropouts under the age of 35 (0.087) and proportion of high school dropouts over the age of 65 (0.129).

Then, I could run the test to see if there was sufficient evidence to conclude there was no difference between the proportions. At the 95% confidence level, p-value > alpha (0.059 > 0.05), and absolute value of zStat is within the critical values (1.89 < 1.96). We fail to reject the null hypothesis, and conclude there is insufficient evidence to suggest a difference between the proportion of people aged 35 or younger who were high school dropouts, and people 65 or older who were high school dropouts.

*work shown in excel file

4. When breaking down the number of respondents with both a particular set of earners and a particular set of children, there are four groups with the same number of respondents. Those groups are:

• EARNRS = 1, CHILDS = 1
• EARNRS = 1, CHILDS = 3
• EARNRS = 2, CHILDS = 0
• EARNRS = 2, CHILDS = 1

First consider the two groups of households with two wage earners (EARNRS = 2, CHILDS = 0 and EARNRS = 2, CHILDS = 1). Is there a di↵erence between the average annual income of these two groups? Initially run your test treating the samples as independent, then rerun your analysis by sorting and matching households by age (it won't be perfect, but will be good enough for you to use). Does your conclusion change when you use matched pairs analysis? What does that tell you about age as a controlling factor?

To determine whether there is a difference in average annual income between households with two wage earners and no children (EARNRS = 2, CHILDS = 0) and households with two wage earners and one child (EARNRS = 2, CHILDS = 1), we first treated the samples as independent.

Part A: Independent Samples

I first ran an F-test to determine equal variances. The p-value from the F-test was approximately 0.060, which is greater than alpha = 0.05. Therefore, we fail to reject the null hypothesis of equal variances and use a two-sample t-test assuming equal variances.

The mean income for the 2 earner/0 child group was approximately $61,660, while the mean income for the 2 earner/1 child group was approximately $66,763. The estimated difference in means was −$5,103, indicating that households with one child earned about $5,103 more on average.

At the 95% confidence level, p-value > alpha (0.11 > 0.05), and the absolute value of tStat being within the critical values (1.57 < 1.97). We fail to reject the null hypothesis and conclude there is insufficient evidence to suggest a difference between incomes of 2 Earners/0 Childs and 2 Earners/1 Childs.

Part B: Matched Pairs (Controlling for Age)

Next, I sorted both groups by age and paired households by similar ages. I then analyzed the income differences within each matched pair using a paired t-test.

The mean difference in income remained approximately −$5,103. The paired t-test produced a test statistic of −1.61 with a two-tailed p-value of 0.110. Because this p-value is greater than 0.05, we again fail to reject the null hypothesis.

At 95% confidence level, p-value > alpha (0.11 > 0.05), therefore we fail to reject the null hypothesis and conclude there is insufficient evidence to suggest a difference between incomes of 2 Earners/0 Childs and 2 Earners/1 Childs.

The conclusion does not change after controlling for age through matched pairs analysis. In both the independent and matched analyses, there is insufficient statistical evidence to suggest a difference in mean income between households.

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5. At the 95% confidence level, are there any differences between the four groups? Make sure you check to make sure you are not violating the assumptions of your test.

Since I am comparing more than 2 samples here (1 Earner,/1 Child, 1 Earner/3 Childs, 2 Earners/0 Child, and 2 Earners/1 Child, I had to use an ANOVA test. This test looked to see if there was any difference between these four households and income.

After running the test, at the 95% confidence level, Fstat < Fcrit (1.26 < 2.62) and p-value > alpha (0.29 > 0.05). Therefore, we fail to reject the null hypothesis and conclude there is insufficient evidence to suggest there is a difference between any of the 4 households.

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